

Day 4 — Indian Mathematics: Specifics

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students engage three specific Indian mathematical works at primary-source level, articulating what each established and what limits each had.

Materials

- Printed handout: one problem from Bhāskara II's *Līlāvātī* in English translation
- Printed handout: statement of Mādhava's series for π

Flow

Warm-up (5 min)

Recall: name 3 Indian mathematical contributions (from Day 6 of Middle Module 1, if students did it; or fresh recall).

Core (25 min)

1. **The Bakhshali manuscript** (~7th c. CE — fragments possibly earlier).
 - Birch-bark manuscript discovered in 1881.
 - Contains the earliest known use of zero as a number with a dot symbol.
 - Word problems on arithmetic, equations.
 - **Why it matters:** documents Indian arithmetic ~500 years before European adoption.
2. **Bhāskara II (1114–1185 CE).** *Līlāvātī* and *Bījagaṇita*.
 - Arithmetic, algebra, geometry, mensuration.
 - Solved Pell's equation (the equation $x^2 - Ny^2 = 1$) centuries before Pell himself (1650s).
 - Introduced cycle methods for indeterminate equations.
 - Read aloud one *Līlāvātī* word problem (e.g. the swan problem).
3. **Mādhava of Saṅgamagrāma (~1340–1425 CE).** Kerala school of mathematics.
 - Infinite series for π : $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$ (now sometimes called Madhava-Leibniz series).
 - Series for sine, cosine, arctan.
 - Came very close to calculus (~250 years before Newton and Leibniz).
 - **What he did NOT do:** did not develop the fundamental theorem of calculus formally; did not unify the series approach with a differential framework.

Discussion (10 min)

- “*Madhava had infinite series 300 years before Leibniz. So why is the series still called ‘Leibniz series’ in most textbooks?*”
- Possible answers: European mathematical literature was the dominant pipeline; Kerala school’s work was not translated to European languages until much later; cite-attribution is conservative.
- The corrective: modern math history is increasingly using “Madhava-Leibniz series” — a real, slow correction.

Wrap-up (5 min)

- Preview Day 5: “*Tomorrow — mid-module + paper topic commitment.*”

Differentiation

- **Tier up:** Compute Mādhava’s series numerically for 100 terms. How close to π ?
- **Tier down:** Focus on Bakhshali + Bhāskara names with dates.

Teacher notes

- The Madhava-Leibniz attribution is one of the most pedagogically useful examples of “real Indian contribution, slow Western recognition.”
- Don’t oversell: Mādhava’s work was discovered/recognised in the West via George Joseph and others in the late 20th c.

Citation(s) used

- Plofker (2009), Chapters 4–7.
- Hayashi (1995) on Bakhshali.
- George Joseph (2011). *The Crest of the Peacock: Non-European Roots of Mathematics*. Princeton.